What's the worst that can happen?



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Today's thoughts

Documentation mistakes are:

- Bad (duh!)
- Costly (important!)
- Worth seeing examples:
  - Car recalls
  - Aviation crashes
  - Potentially exploding bombs

Every time you make a typo

The errorists win



Mistakes are bad



Mistakes are bugs





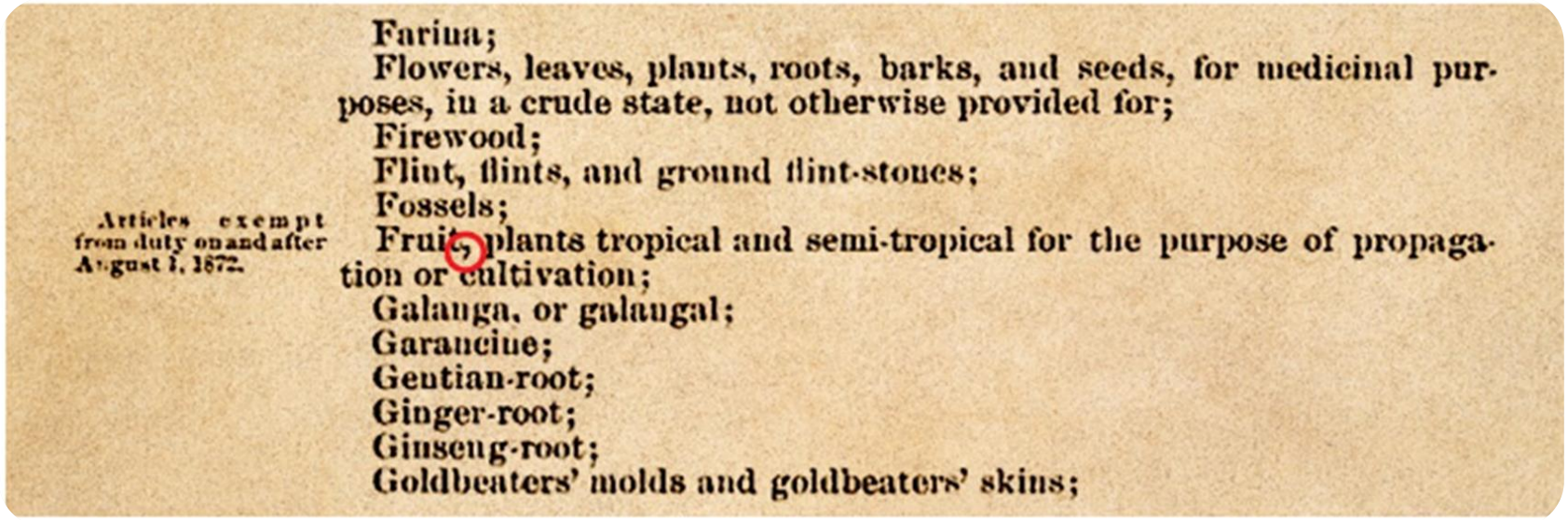
- Harder to spot
- More difficult to eliminate
- Results not immediately felt







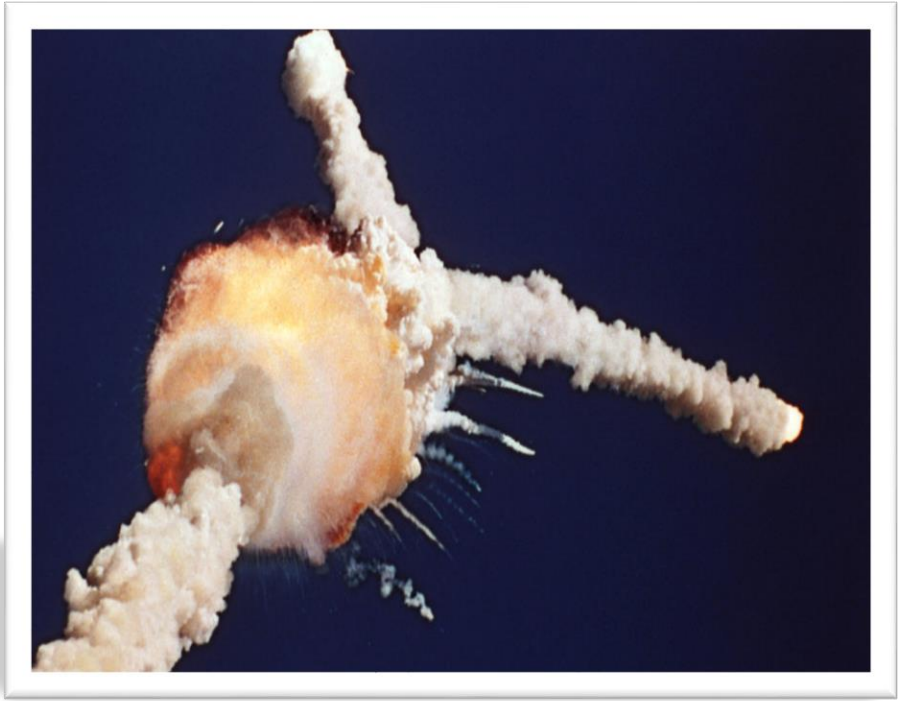
One of the oldest tax loopholes



Articles exempt from duty on and after August 1, 1872.

Farina;
Flowers, leaves, plants, roots, barks, and seeds, for medicinal purposes, in a crude state, not otherwise provided for;
Firewood;
Flint, flints, and ground flint-stones;
Fossels;
Fruit, plants tropical and semi-tropical for the purpose of propagation or cultivation;
Galanga, or galangal;
Garancine;
Gentian-root;
Ginger-root;
Ginseng-root;
Goldbeaters' molds and goldbeaters' skins;

Most expensive hyphen in history



Mariner 1 aborted because of this mistake

$$\dot{\bar{R}}_n$$

610,000 shares

1 yen

340,000,000 USD



The most dangerous profession in the world

Documentation mistakes are dangerous

- Car manuals
- Medicine instructions
- Military hardware maintenance manuals



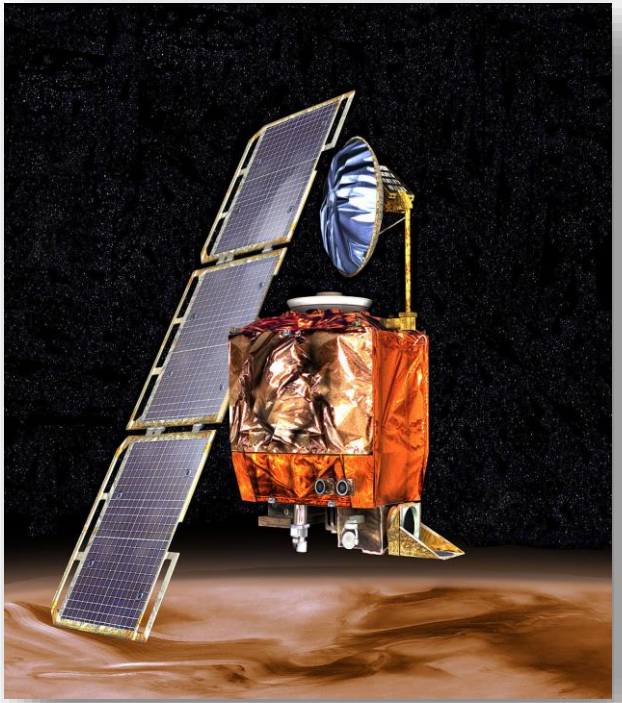
Over 3,000 cars recalled

- “Inaccurate description of the actual operation of the ‘Front Passenger Occupant Classification System.’”
- Sections on airbags needed revision



Mars Climate Orbiter exploded

- Lockheed Martin didn't read the software specifications
- NASA failed to review Lockheed Martin software





$327,600,000

Whoa! That's a big number!

An IDF helicopter crashed

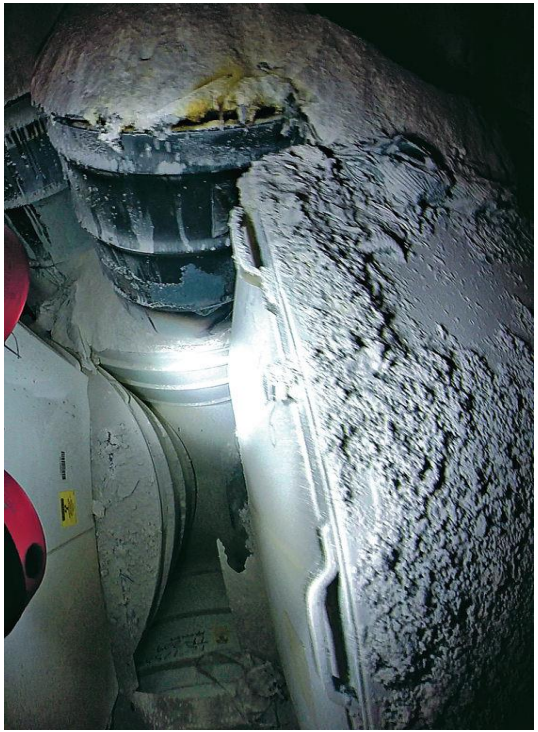
- Steering rod incorrectly installed
- Maintenance manual missing a step



No this isn’t what happened



A drum of radioactive waste leaked



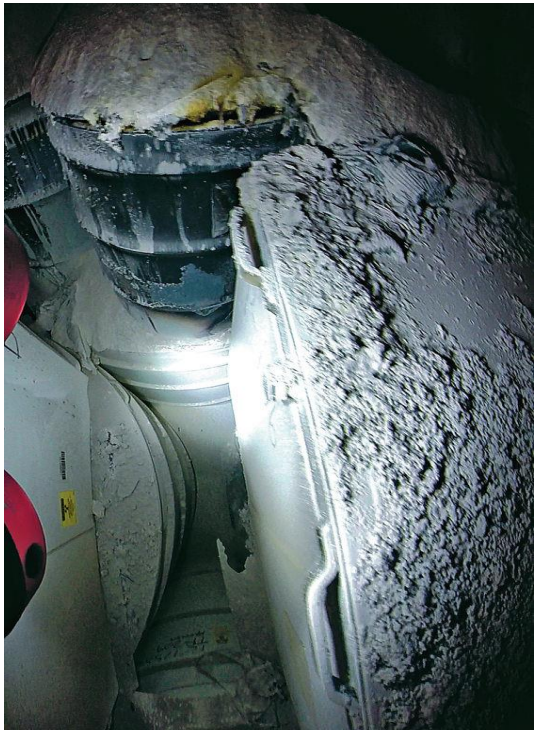
- **1,600˚ – Fahrenheit**
- **20 people – Contaminated**
- **3 years – Plant shut down**



A drum of radioactive waste leaked

Because of a typo!!!

“organic” vs. “inorganic”





$500,000,000

Whoa! That's an even bigger number!



Conclusions

- Everyone makes mistakes
- Mistakes are bad
- Mistakes can be very costly

How do we catch them?

Possible solutions



- SME reviews – lots of them
- Tech writer access to SME content
- Sufficiently detailed procedures
- Better communication – writers and reviewers
- Enforced corporate style guide
- Documentation QA

Thanks!

Any questions?

You can contact me at:

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Credits



- 1872 Tariff act
  - “The High Cost of Small Mistakes: The Most Expensive Typos of All Time” https://www.six-degrees.com/the-high-cost-of-small-mistakes-the-most-expensive-typos-of-all-time/
- Mariner 1 launch failure
  - Wikipedia https://en.wikipedia.org/wiki/Mariner_1
- Mizuho Securities
  - “Stock Trade Typo Costs Firm $225M” https://www.cbsnews.com/news/stock-trade-typo-costs-firm-225m/
  - “Fat finger costs Japanese broker US$225m” http://www.chinadaily.com.cn/english/doc/2005-12/10/content_502260.htm

Credits



- Toyota and Subaru recall
  - “Scion FR-S and Subaru BRZ Are Recalled for Error in Owner’s Manuals” https://wheels.blogs.nytimes.com/2012/06/07/toyota-recalls-1100-scion-fr-s-sports-cars-for-error-in-owners-manual/
- Mars Orbiter Crash
  - Wikipedia https://en.wikipedia.org/wiki/Mars_Climate_Orbiter#Cause_of_failure
- Helicopter crash
  - “IDF confirms: Steering rod incorrectly installed, causing deadly chopper crash” https://www.timesofisrael.com/idf-confirms-steering-rod-incorrectly-installed-causing-deadly-chopper-crash/

Credits



- Radioactive Bomb
  - "How a Typo May Have Turned a Drum of Radioactive Waste Into a Bomb" https://gizmodo.com/how-a-simple-typo-may-have-turned-a-drum-of-radioactive-1662683094
  - "LANL officials downplayed waste's dangers even after leak" http://www.santafenewmexican.com/special_reports/from_lanl_to_leak/lanl-officials-downplayed-waste-s-dangers-even-after-leak/article_54d7f3d2-8c99-5793-8c17-c4bdb0b72ef1.html?mode=jqm
  - "MANAGEMENT ALERT: Remediation of Selected Transuranic Waste Drums at Los Alamos National Laboratory – Potential Impact on the Shutdown of the Department's Waste Isolation Plant" https://www.energy.gov/sites/prod/files/2014/10/f18/DOE-IG-0922.pdf